Jayakrishna Konda Data Scientist (GenAI & ML)

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SUMMARY

Data Scientist and ML Engineer with 4+ years of experience building and scaling production-grade ML and GenAI systems that improve automation, decision quality, and business outcomes. Specialist in NLP, RAG, and agentic AI, delivering high-performance pipelines through distributed data processing, MLOps, and cloud-native deployment. Proven at translating business problems into scalable AI solutions using Python, PySpark, PyTorch/TensorFlow, Docker, Kubernetes, and AWS optimized for reliability, cost, and impact.

SKILLS

Programming & ML Foundations: Python, SQL, PySpark, Pandas, NumPy, Regression, Classification, Clustering, Feature Engineering, Statistical Analysis, Predictive Modeling, Forecasting, Model Evaluation & Validation

Deep Learning, NLP & GenAI: TensorFlow, PyTorch, scikit-learn, NLP, LLMs, Prompt Engineering, RAG, Embeddings, Semantic Search, Text Classification, Fine-Tuning, Agentic AI Systems

LLM Frameworks & Orchestration: LangChain, LangGraph, Tool Calling, Multi-Step Reasoning, Workflow Orchestration

MLOps & Model Lifecycle: CI/CD, Model Versioning, Automated Retraining, Experiment Tracking, A/B Testing, Human-in-the-Loop Evaluation, Monitoring & Drift Detection

Big Data & Distributed Systems: Apache Spark (PySpark), Batch & Real-Time Inference, Large-Scale Feature Engineering, Distributed Data Processing

Cloud & Deployment: AWS (Lambda, S3, Step Functions, DynamoDB, Redshift), Docker, Kubernetes, Cloud-Native Microservices, Event-Driven Architectures, RESTful APIs

Performance Optimization: GPU Training & Inference, CUDA, Cost-Aware ML Systems

EXPERIENCE

Data Scientist (GenAI & ML) | Enigma Technologies **Jan 2025 - Present | Remote, USA**

- Design and deploy NLP and GenAI solutions using prompt engineering, RAG, and fine-tuning to automate large-scale text understanding workflows, reducing manual analysis effort by 30%+ and accelerating business decision-making.
- Build and deploy end-to-end ML/DL models for regression, classification, and clustering, leveraging statistical analysis, predictive modeling, and forecasting to improve accuracy, reliability, and operational decision quality by 20% in production environments.
- Partner with product and business stakeholders to translate analytical and GenAI requirements into scalable ML systems, delivering agentic AI workflows with LangChain and LangGraph that enable faster, more consistent outcomes.
- Establish robust ML and LLM evaluation frameworks, combining automated metrics, human-in-the-loop review, and live A/B testing to continuously enhance output relevance, quality, safety, and model trustworthiness.
- Deliver production-grade ML platforms using Docker, Kubernetes, and CI/CD pipelines, ensuring reproducible training, reliable deployments, monitoring, and horizontally scalable inference across cloud environments while optimizing compute costs.
- Optimize large-scale training and inference workflows using GPU-enabled infrastructure, CUDA acceleration, and GPU-optimized libraries, improving throughput and latency while maintaining system reliability and operational efficiency.

Machine Learning Engineer | Cognizant **Jan 2020 - Jul 2023 | India**

- Delivered production-grade NLP systems for context extraction, semantic search, and text classification by automating large-scale unstructured data workflows across multiple pipelines, resulting in a 35% reduction in manual review effort while improving SLAs.
- Architected scalable machine learning pipelines supporting end-to-end model development, including training, evaluation, and batch/real-time inference across multi-million–record datasets, leveraging Python, scikit-learn, PySpark, TensorFlow, and PyTorch.
- Established enterprise-grade MLOps practices across the full model lifecycle by implementing CI/CD pipelines, model versioning, automated retraining, and performance monitoring, cutting model release cycles by 30% and improving production stability.
- Engineered cloud-native inference services using AWS Lambda, microservices, and event-driven architectures, enabling low-latency and cost-efficient model serving that reduced inference latency by 40% while optimizing infrastructure and operational expenses.
- Optimized distributed PySpark workflows for large-scale NLP feature extraction and batch inference by designing fault-tolerant data processing pipelines, reliably handling 1M+ text records per day with consistent throughput and high availability.
- Implemented retrieval-based and context-aware NLP pipelines using embeddings and semantic indexing to enhance semantic understanding, significantly improving search relevance, classification accuracy, and downstream system effectiveness.
- Authored technical documentation, reproducible pipelines, and internal best practices, streamlining onboarding and knowledge transfer across ML and data teams.

PROJECTS

Self-Hosted AI/ML Infrastructure (Batcave)

- Designed and deployed a containerized AI platform running local LLMs and embedding models, enabling secure, end-to-end RAG pipelines and semantic search across 200+ page unstructured documents using transformer embeddings and FAISS.
- Optimized and productionized inference and retrieval workflows by refactoring pipelines and tuning container resources, reducing latency while implementing reproducible, production-grade deployments with Docker, reverse proxy routing, and HTTPS security.

Financial Crisis Analysis Tool

- Implemented FinBERT-based sentiment analysis achieving 87% accuracy across 10,000+ financial news articles, enabling scalable text analytics, model evaluation, and data-driven insights for financial decision-making workflows and automation.
- Executed 1,000+ Monte Carlo simulations to analyze portfolio risk with 95% confidence intervals and calculated CAPM metrics for 50+ stocks, automating alpha/beta analysis and reducing effort by 80%.

RAG-powered Podcast Generator (Capstone)

- Built AI engine processing 200+ page PDFs with 98% OCR accuracy, enabling character-voiced audio conversion, multi-voice narration, and 90% character identification accuracy across diverse literary genres.
- Integrated LLM fine-tuning for narrative tone consistency across genres and optimized LangChain pipeline with local Ollama deployment, achieving 4x faster processing performance and scalable end-to-end audio generation workflows.

EDUCATION

Master of Science in Data Science — **Aug 2023 – Jun 2025**
University of Maryland, Baltimore County — Baltimore, MD

Bachelors in Computer Science and Engineering — **Aug 2015 – Aug 2019**
Sreenidhi Institute of Science and Technology — Hyderabad, India